Outward mutation in Nimboran

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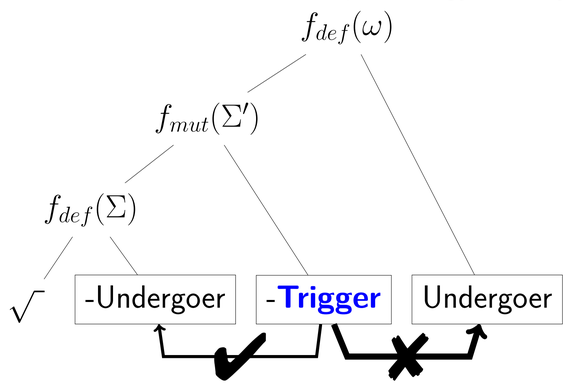
1. Strict Base Mutation

- Cyclic non-morphemic Approaches to mutation, e.g. Output-Output Anti-Faithfulness (OOA) (Alderete, 1999), Cophonology Theory (CT) (Inkelas, 1998)

  Prediction: Strict Base Mutation, i.e. no outward mutation.

  Mutation: Morphologically conditioned changes in segment quality.
- CT: domain of the mutation cophonology triggered by an affix includes only affix and base.

(1) No outward mutation in Cophonology Theory



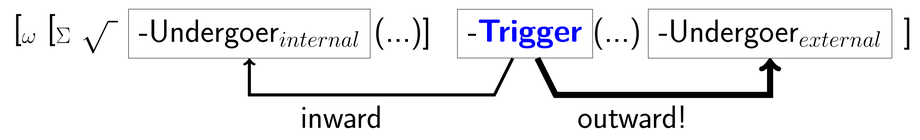
2. Main Claim

- Nimboran outward vowel mutation → counterexample to Strict Base Mutation.
- In contrast to previous cases (Trommer & Zimmermann, 2015b,a), bidirectional mutation.

  Claim: Non-cyclic computation and specification of mutation domains neccesary.

  Proposal: Analysis with attachment of featural affixes calculated in an OT system.

(2) Structure of Nimboran vowel mutation



3. Data: Nimboran Vowel Mutation

- Nimboran (isolate, Indonesia) vowel mutation cosists of two parts (cf. Inkelas 1993).
- If trigger present ,e.g. *-ta* DUR, low vowels in non-final affixes are fronted (3a).
- Final vowels are raised to unrounded high vowels (3a).
- Underlying vowels occur in absence of a trigger (3b).

(3) '-ta' DUR triggers mutation (Anceaux, 1965, 186,234)

a. /ŋgedóu-**ta**-man-t-ám/ → [ŋgedúo-te-men-t-ím]
draw-DUR-INCL.DU.S-PRS-INCL
'You (sg) and I are drawing here.'

b. /ŋgedúo-man-t-ám/ → [ŋgedúo-man-t-ám]
draw.SG-INCL.DU.S-PRS-INCL
'You (sg) and I draw here.'

- **Inner affixes trigger mutation on outer affixes**.
- Mutation can also go in both directions (4).

(4) Bidirectional mutation (Anceaux, 1965, 188)
/ŋgedúo-man-**sa**-t-ám/ → [ŋgedúo-men-se-t-ím]
draw.SG-INCL.DU.S-7.LOC-PRS-INCL
'You and I will draw from here to below.'

- Non-final mutation affects multiple vowels.

(5) Iterative mutation (Anceaux, 1965, 236)
/ŋgedóu-**ta**-ma-na-ŋga-t-ám/ → [ŋgedóu-te-me-ne-ŋge-t-ím]
draw-DUR-INCL.DU.S-ITER-5LOC-PRS-INCL
'You (sg) and I are drawing repeatedly far away.'

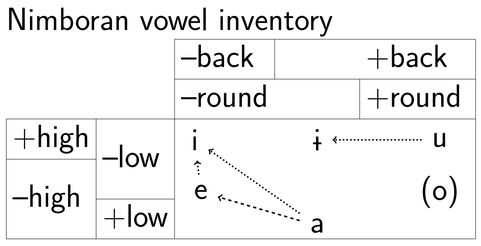
- Final vowel mutation affects only the last vowel.

(6) Final vowel mutation (Anceaux, 1965, 126,141)

a. /sá-**n**-d-u/ → [sá-n-d-ɨ]
drive.away.SG-PART-FUT-1
'I will drive away from here.'

b. /iá-k-**mam**-bá-e/ → [iá-ke-mem-bé-i]
ask-DU.S-PART-2LOC-2
'You two ask him above.'

(7) Nimboran vowel inventory



Dotted: final mutation

Dashed: non-final mutation

4. Assumptions

Crucial:

- Generalized Non-Linear Affixation: Mutation is triggered by affixed **floating phonological features** (Bermúdez-Otero, 2012).
- **Affix Pivots**: Affixes are specified for (prominent) attachment position (Yu, 2007).
- Foot structure: Affixes are a separate stress domain (=foot) in Nimboran (cf. Inkelas 1993).

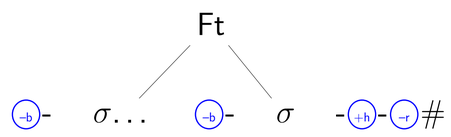
Additional:

- Radical Containment OT: no deletion, only insertion and marking of association lines (Trommer, 2011).
- Morphological Colors: Phonology is sensitive to identity of morphological affiliation (van Oostendorp, 2007; Revithiadou, 2007).
- Overwriting as Contiguity: A morphological CONTIGUITY constraint (cf. Landman 2002; Finley 2009) forces featural affixes to overwrite intervening features on the same tier (cf. Trommer 2015).

5. Analysis: Floating Features and Trigger Constraints

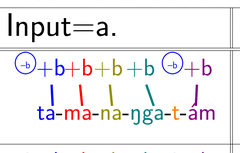
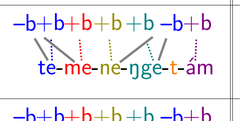
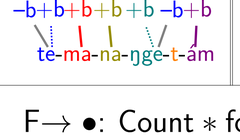
- Featural affixes attach at several positions in the word, delimiting the domains of mutation.
- General phonological constraints govern docking of features.

(9) Featural affixes in Nimboran mutation



- Non-final mutation is analyzed as **two floating [-back] features**, attaching before the last foot and before the last syllable.
- CONTIGUITY prefers the candidate where all **interveners are overwritten**.
- Simple feature docking cannot satisfy this constraint.

(10) Non-final vowel mutation by CONTIGUITY

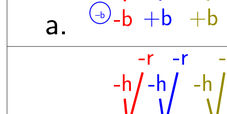
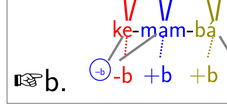
Input=a.	CONT(bk)	bk→•
a. 	*!	**
☞b. 		****
c. 	*!	**

(11) F→ •: Count * for every feature F that is not visibly associated to a segmental root node.

(12) CONTIGUITY(back): Count one violation for every segmental root node $\bullet_1$, such that $\bullet_1$ is visibly associated to a [±back] feature b_1 of color c_1 that intervenes between two [±back] features of the same color.

- Final mutation involves separate triggers.
- **Suffixed** -(+h),-(-r) allow for **markedness reduction**.
- One of the features is associated if it allows to delink marked features.
- Vacuous docking is blocked by the TWINSISTERCONDITION.
- Faithfulness violations are only allowed if two feature values compete for association.

(13) Final vowel mutation allows markedness reduction

Input=a.	TSC	rd→•	hi→•	*+rd	*-hi
a. 		*	*	*!	*
☞b. 		*	*		*
c. 	*!	*		*	*

(14) Constraint for final vowel mutation

a. *F: Count for every segmental root node visibly associated to a F feature.

b. TWINSISTERCONDITION: Count * for each segmental root node that is associated to the same feature twice (cf. Clements & Keyser 1983).

(15) Both mutations combined

Input=a.	CONT(bk)	TSC	rd→•	hi→•	*–hi
a.	*!		*	*	****
☞b.		*	*	*	***

6. Discussion

- Inner affixes can trigger mutation on outer affixes.
- This is a strong argument against cyclic non-morphemic approaches to mutation.
- OOA requires a difference relative to a base.
- CT restricts mutation to the base.
- Outer affixes are not part of the base and thus should not be included in the domain of mutation.
- An alternative morphological analysis, taking the affix domain as a separate stem, still fails, because of the bidirectionality.
- Generalized Non-Linear Affixation derives the specification of the domains of mutation by lexical information of attachment pivots.
- Markedness constraints and CONTIGUITY allow non-cyclic docking of the floating features and overwriting.

Selected References: •Alderete, John. 1999. *Morphologically-governed accent in Optimality Theory*. Amherst: University of Massachesetts dissertation. •Anceaux, J.C. 1965. *The Nimboran language: Phonology and morphology*. 's Gravenhage: Martinus Nijhoff. •Bermúdez-Otero, Ricardo. 2012. The architecture of grammar and the division of labour in exponence. In Jochen Trommer (ed.), *The morphology and phonology of exponence*, 8–83. Oxford: Oxford University Press. •Inkelas, Sharon. 1998. The theoretical status of morphologically conditioned phonology. a case study of dominance effects. In Geert Booij & Jaap van Marle (eds.), Yearbook of morphology 1997, 121–155. Dordrecht: Springer. •Yu, Alan. 2007. A natural history of infixation. Oxford: Oxford University Press.